🎓 EduTrack LMS – Project Report (Final Version)

A Full-Stack Learning Management System (LMS)

---

1. Project Information

- **Project Title**: EduTrack LMS
- **Team Members**:
  - Satyam Kumar
  - Kavya Saraswat
  - Mausam Kumar Dwivedi
  - Rachit Singh
  - Himank Kaushik
- **Repository**: https://github.com/mausam-005/edutrack-lms
- **See the project**: https://edutrack-lms.netlify.app

---

2. Abstract

EduTrack LMS is a scalable and feature-rich Learning Management System designed to modernize educational workflows. Built using the MERN stack with TypeScript, the platform integrates course management, student engagement, and automated assessments into a unified ecosystem.

Unlike traditional LMS platforms that often suffer from complexity or limited flexibility, EduTrack focuses on **usability, modularity, and real-time interaction**. It incorporates role-based dashboards, dynamic content rendering, and event-driven notifications, providing a seamless and responsive learning experience across devices.

---

3. Introduction

3.1 Background

The rapid digitalization of education has transformed how knowledge is delivered and consumed. Institutions increasingly require platforms that are not only functional but also scalable, interactive, and user-friendly.

3.2 Problem Statement

Existing educational systems face several challenges:

- Fragmented data management across platforms
- Lack of real-time communication between stakeholders
- Limited automation in assessments and progress tracking
- Overly complex systems unsuitable for smaller institutions

3.3 Objectives

- Develop a **centralized learning platform**
- Implement a **secure role-based authentication system**
- Provide **interactive quizzes with automated evaluation**
- Ensure a **responsive and accessible user interface**
- Apply **industry-standard architecture and design principles**

---

4. System Requirements

4.1 Technology Stack

Layer	Technology
Frontend	React 18, TypeScript, Tailwind CSS, Vite
Backend	Node.js, Express.js (TypeScript)
Database	MongoDB Atlas (Mongoose ODM)
Authentication	JWT, bcrypt.js
Deployment	Netlify (Frontend), Render (Backend)

4.2 Functional Requirements

- **User Authentication**
  Secure login and registration with role-based access (Student, Teacher, Admin)
- **Course Management**
  Teachers can create, update, and manage courses; students can browse and enroll
- **Study Materials**
  Structured delivery of learning resources including text, PDFs, and videos
- **Quiz System**
  Timed MCQ-based quizzes with automated evaluation and instant results
- **Notification System**
  Real-time alerts for enrollments and course updates
- **Dashboards**
  Role-specific analytics and activity summaries

4.3 Non-Functional Requirements

- **Scalability** → Layered architecture supports growth
- **Security** → JWT-based authentication and encrypted storage
- **Performance** → Optimized queries and efficient rendering
- **Responsiveness** → Mobile-first UI design
- **Maintainability** → Modular and clean codebase

---

5. System Design

5.1 Architecture Overview

EduTrack follows a **Layered (N-Tier) Architecture**:

1. **Presentation Layer** → React-based user interface
2. **Routing Layer** → Express.js APIs with middleware
3. **Business Logic Layer** → Services handling application logic

4. **Data Access Layer** → MongoDB with Mongoose models

This separation ensures **scalability, maintainability, and flexibility**.

5.2 Database Design

- **User** → Stores user credentials and roles
- **Course** → Contains course details and instructor references
- **Enrollment** → Tracks student participation
- **Quiz** → Stores MCQs and evaluation rules
- **Result** → Maintains performance records
- **Notification** → Handles system alerts

---

6. Implementation

6.1 Design Patterns Used

- **Singleton Pattern** → Single database connection
- **MVC Architecture** → Separation of concerns
- **Observer Pattern** → Event-driven notification system
- **SOLID Principles** → Clean and maintainable structure

6.2 Core Feature Implementation

- **JWT Middleware**
  Ensures secure access and role-based authorization

- **Dynamic Rendering**
  React state enables smooth UI transitions

- **Responsive Design**
  Tailwind CSS ensures cross-device compatibility

- **Event-Based Notifications**
  Decoupled communication between modules

---

7. Results and User Interface

- Clean and validated authentication pages
- Student dashboard with progress tracking
- Teacher dashboard with course analytics
- Minimal and focused course viewer
- Light/Dark mode for accessibility

---

8. Conclusion

EduTrack LMS successfully delivers a **robust, scalable, and user-friendly platform** for digital education. By combining modern technologies with structured architectural practices, the system addresses key limitations of traditional learning solutions.

The project demonstrates strong implementation of **full-stack development, design patterns, and system design principles**, making it suitable for real-world deployment.

---

9. References

- React Documentation – https://react.dev
- MongoDB Mongoose – https://mongoosejs.com
- Express.js Documentation
- Tailwind CSS – https://tailwindcss.com
- *Design Patterns: Elements of Reusable Object-Oriented Software* – Gamma et al.